

강의계획서

2026학년도 1학기

과목명	프로그래밍과 문제해결		교과목 번호	MC3203	학점	3	강의실	창708
			교과 구분	AP 과목	시수	3		
담당교원	이름	분반	Office	Office Hour*	연락처		이메일	
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1. 교과목 개요

Course Objectives

On completion of this course, you'll hopefully be able to:

- Use the Python programming language
 - Programming is needed in all areas of science and engineering, although very different languages are used.
- Think computationally to solve simple algorithmic problems
 - Computational thinking influences how you go about solving a problem

Contents

- Python programming language
 - lists, tuples, sets, dictionaries, strings, files
 - high-order functions
 - sorting with user-defined orders
 - recursion, fast recursion with memoization
 - user-defined objects, operator overloading
 - encapsulation, inheritance, polymorphism
 - memory model for objects
- Problem solving using Python
 - scientific computation (e.g. N-body simulation, Gaussian elimination)
 - graph traversals (BFS/DFS) & applications (e.g. maze robot, 8-puzzle, flood fill)
 - image processing, recursive graphics
 - recursive puzzles (e.g. N-Queen, Sudoku, minesweeper)
 - winning strategy for simple combinatorial games (e.g. Tic-Tac-Toe)
 - search heuristics for combinatorial games (e.g. α - β pruning for Chess)
 - basic geometric problems (e.g. plane sweeping, orientation, convex hull)

2. 중점 목표 역량

교과 핵심 역량			
problem-solving skills	logical thinking	problem discovery and definition	creativity

3. 교재 및 참고도서

가. 주교재 : 자체 제작 수업 자료

4. 수업 방법 :

가. 강의 및 실습

5. 평가 기준 및 방법

가. 평가 기준

수업 중 과제	중간고사	기말고사	출결 및 태도	합계
30%	10%	50%	10%	100%

나. 평가 방법

- Programming assignment for each class
- Midterm exam: written or practical test (2 hrs)
- Final exam: practical test (3-4 hrs)
- Make-up final exam for officially approved absence (공결/병결)
 - 13:00 on the first day when the reasons for all absentees are resolved
(모든 공결/병결자의 사유가 해소되는 첫날 13:00)
 - 0 point if do not show up on the make-up final test
 - no make-up midterm exam

6. 학점 부여 기준

- 출결 및 태도: 총 수업 시간의 3/4 이상 부여시 100% 부여
- 병결/공결 등으로 정규 기말고사에 미응시 경우: 대체시험 실시하여 그대로 기말고사 점수로 반영
- 학점 부여 기준: KAIST/POSTECH AP 담당 교수의 피드백으로 결정
- 과목우수상 기준: 원점수 기준으로 정렬하여 결정

7. 주별 강의 계획

주(기간)	주제 및 내용		(비고)
제1주 2.23.~3.1.	CS1/CS2 review		
핵심역량	problem-solving skills, logical thinking		
제2주 3.2.~3.8.	Recursion I	inductive definition of function	
	Recursion II	combinatorial game, winning strategy	
핵심역량	problem-solving skills, logical thinking		
제3주 3.9.~3.15.	High-order functions	function as object, anonymous function	
	More on functions	scope, default/named param, var-len arg	
핵심역량	problem-solving skills, logical thinking		
제4주 3.16.~3.22.	Lists I	basic functionalities of lists	
	Lists II	sorting with user-defined order relation	
핵심역량	problem-solving skills, logical thinking		
제5주 3.23.~3.29.	Tuples/Strings	basic functionalities of tuples/strings	
	Sets/Files	basic functionalities of sets/files	
	Dictionaries	basic functionalities of dictionaries	
핵심역량	problem-solving skills, logical thinking		
제6주 3.30.~4.5.	User-defined objects I overloadable built-in operators		
핵심역량	problem-solving skills, logical thinking		
제7주 4.6.~4.12.	Recursion III	memoization with dictionaries (DP)	
	Memory layout for lists/objects	alias, shallow/deep copy/equality	
핵심역량	problem-solving skills, logical thinking		
제8주 4.13.~4.19.	중간고사 (4.14~4.17.)		

주(기간)	주제 및 내용		(비고)
제9주 4.20.~4.26.	Graphics & GUI recursive graphics, N-Queen/Sudoku		
핵심역량	problem-solving skills, logical thinking		
제10주 4.27.~5.3.	Image processing	basic operations for 2D images, flood fill	
핵심역량	problem discovery and definition, creativity		
제11주 5.4.~5.10.	Graphs Graph traversals (DFS/BFS)	basic notions, adjacency matrix/list applications (e.g. maze robot, 8-puzzle)	5.5.(화) 어린이날
핵심역량	problem discovery and definition, creativity		
제12주 5.11.~5.17.	User-defined objects II N-body simulation	class variables/methods, encapsulation N-body moving under gravitational force	
핵심역량	problem discovery and definition, creativity		
제13주 5.18.~5.24.	User-defined objects III Search heuristics for comb. games Game-tree search for chess	inheritance, polymorphism α - β pruning, Monte-Carlo tree-search α - β search for chess, evaluation heuristic	
핵심역량	problem discovery and definition, creativity		
제14주 5.25.~5.31.	Basic geometric problems plane sweeping, CCW, convex hull		5.25.(월) 대체공휴일
핵심역량	problem-solving skills, logical thinking		
제15주 6.1.~6.7.	Wrap-up for the final exam		6.3.(화) 지방선거일
핵심역량	problem-solving skills, logical thinking		
제16주 6.8.~6.14.	보강 주간		
제17주 6.15.~6.19.	기말고사 (6.18~6.19.)		

※ 학사 일정에 따라 강의 계획이 변경될 수 있음